Fundamentals Of Data Science

Reflective Writing

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The FODS final project was something I genuinely looked forward to, and working on the Student Management System turned out to be more rewarding than I had expected. Our vision was clear, to replace a slow and error prone manual record keeping process with a digital and automated system. But what I didn't expect was how much I would learn beyond just writing code. It taught me how the data, when handled well, can actually help people in making better decisions. My main role in the project was building the analytics dashboard. This included creating grade trend charts using Matplotlib, running a correlation analysis between ECA participation and academic performance, and setting up performance alerts for students scoring below 70%. Alongside that, I contributed to writing the final report and formatting the documentation.

At first, I wasn't fully confident about handling the data visualization part on my own. I had some experience with Matplotlib, but making the charts genuinely useful and not just visually present was harder than I expected. Getting the correlation graph to tell a clear story took several attempts. When it finally came together, it was really satisfying to see those raw data turn into something meaningful. Writing the final report also surprised me as I enjoyed it more than I thought I would. Having to explain the system in words, describing the DFD levels, the architecture, and how everything connected, helped me to understand our own project more deeply. It made me see the system from a wider view, not just from the part I built.

Looking at what went well, the teamwork was one of our biggest strengths. Because everyone had a defined role, there was very little confusion or overlap, and we were able to support each other when things got difficult. On the technical side, I think the analytics dashboard added real value to the project. It wasn't just about displaying numbers, it was about making those numbers useful. For instance, the performance alert system could genuinely help a module instructor to identify the students who needed extra support. That sense of purpose made the work feel more than just an assignment.

However, I also had to deal with challenges of keeping the data consistent across multiple text files as any mistake in one place could affect everything else. I understood this more deeply while working on the analytics, because the dashboard depended on all those files being accurate. It showed me how interconnected every part of the system really was.
I was also unable to manage my time properly as I spent too long in the middle stage figuring out the visualization logic, and that created pressure toward the end.

In the future, I will start working earlier and beak down my task into smaller and clearer steps, and ask for input earlier rather than working in isolation, to make my working flow smooth. To address the data consistency challenge I faced, I will look into learning how to use proper databases in future projects, so that managing and syncing data becomes more reliable and less error-prone than working across multiple text files. I will also dive deeper

into tools like Pandas and NumPy, as I only used them in this project at a basic level. Additionally, I want to explore building dashboards in a proper GUI or web-based format, since the CLI has its limits when it comes to presenting data.